

Instruction Pipelining Principles

Q. What is Instruction Pipelining? Explain the stages of a standard instruction pipeline, its working mechanism, and its advantages.

****Definition to Remember****

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1. Stages of a Standard Pipeline

A classic CPU pipeline divides instruction execution into four distinct stages:

1. Fetch

2. Working Mechanism

* ****Without Pipelining (Sequential):**** The CPU finishes all 4 stages of Instruction 1 before starting Instruction 2. (3 instructions take 12 clock cycles).

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3. Pipeline Timing Diagram

text

Clock C

4. Advantages of Pipelining

* ****Increased Throughput:**** Drastically increases the number of instructions executed per second. (Ideal CPI approaches 1).

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****Must-Write Points (for 10 marks)****

1. ****Inst**

*****Quick Recall*****

***`Pipeline`
(Stagge***